

BERRY-ESSEEN RESULTS FOR THE TRIMMED MEAN

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SUMMARY

A Berry-Esseen rate of $O(n^{-1/2})$ is proved for the trimmed mean giving improvements on those existing in the literature.

1. Introduction

In the large sample theory a large variety of tools are currently available for calculating the asymptotic distributions of statistics. However, establishing a rate at which such asymptotic distributions are approached is a comparatively underdeveloped area. Some of the major tools presently available for this purpose are

- (a) Reducing the statistic in some way or other to a sum of i.i.d. random variables to which a well-known theorem of Berry (1941) and Esseen (1945) can be applied. This approach usually yields a rate of $O(n^{-1/2})$.
- (b) If the statistic can be written as a sum of i.i.d. random variables plus a remainder term, converging to zero in probability, the approach of (a) can be applied. However we usually obtain a rate worse than $O(n^{-1/2})$ by this method. See e.g. De Wet (1974).
- (c) The Skorokhod embedding (see e.g. Skorokhod (1965)) has recently been applied with success to statistics which are in some way functionals of sums of independent random variables. See e.g. Rosenkrantz (1967), (1969), Kennedy (1972) and Sawyer (1972). Rates obtained by this method are usually worse than $O(n^{-1/2})$; in many cases $O(n^{-1/4})$ is the best obtainable rate by this method.
- (d) A new characterization of a normal random variable has recently been used by Stein (1972) to obtain rates of convergence of $O(n^{-1/2})$ for sums of dependent random variables. See also Erickson (1974) for an application of Stein's technique.

In this paper we use methods (a) and (b) to obtain Berry-Esseen results for the trimmed mean. We obtain rates of $O(n^{-1/2})$ and $O(n^{-1/2}(\log n)^2)$ which are improvements on those of Rosenkrantz and O'Reilly (1972) who obtained $O(n^{-1/4} \log n)$ and De Wet (1974) who obtained $O(n^{-1/2}(\log n)^2)$. We note in passing that our methods of proof can also be used to obtain analogous results for the Winsorized mean.

Let $X_{1n}, X_{2n}, \dots, X_{nn}$ be the order statistics of an independent sample of size n from a population having distribution function (df.) F . Let (in this section) $0 < \alpha < 1/2$ be fixed and put $r = [\alpha n]$, $s = [(1 - \alpha)n]$ and $\gamma = 1 - 2\alpha$. Let g_1 and g_2 be the (unique) α and $1 - \alpha$ quantiles of F , i.e. $F(g_1) = \alpha$, $F(g_2) = 1 - \alpha$. We assume throughout the paper that F has at least two derivatives in neighbourhoods of g_1 and g_2 with F'' bounded in these neighbourhoods while $f_1 \equiv F'(g_1) = f(g_1)$ and $f_2 \equiv F'(g_2) = f(g_2)$ are positive.

Define the trimmed mean as

$$(1) \quad M_n = \gamma^{-1} n^{-1} \sum_{i=r+1}^s X_{in}$$

It is well known (see e.g. Stigler (1973)) that the above conditions on F ensure the asymptotic normality of M_n (actually the uniqueness of g_1 and g_2 are sufficient). In this paper we obtain rates for the convergence of the df. of M_n (suitably normalised) to the normal df..

In section 2 we prove a number of preliminaries and in section 3 we present the main results.

2. Preliminary Results

Throughout this section and the next one, c_0 is a fixed, positive constant such that

$$(2) \quad c_0 < (\alpha\lambda)^{-1/2} f_1 \min(F^{-1}(3\alpha/2) - g_1, g_1 - F^{-1}(\alpha/2), F^{-1}((1+\alpha)/2) - g_1, \\ g_1 - F^{-1}((3\alpha-1)/2))$$

with $\lambda = 1 - \alpha$, where we interpret $F^{-1}(y)$ as $+\infty$ for $y \geq 1$ and as $-\infty$ for $y \leq 0$. In this section we do not assume that $0 < \alpha < 1/2$ but only that $0 < \alpha < 1$. Furthermore, we choose c_0 so small that F'' is bounded in $J_0 = (g_1 - c_0(\alpha\lambda)^{1/2}/f_1, g_1 + c_0(\alpha\lambda)^{1/2}/f_1)$. Write $\|f\| = \sup_u f(u)$ and $\|f'\| = \sup_u |f'(u)|$, where in both cases the supremum is taken over J_0 .

c will denote a constant $0 < c < c_0$, and in some cases additional restrictions will be placed on c . We also denote $cn^{1/2}$ by c_n .

Lemma 1

For $a > 0$ and all $b > 0$ we have

$$(3) \quad P(n^{1/2}|X_m - g_1| \geq a \log n) = O(n^{-b})$$

$$(4) \quad P(n^{1/2}|X_{sn} - g_2| \geq a \log n) = O(n^{-b})$$

Proof

Put

$$q_n = P(n^{1/2}(X_m - g_1) \geq a \log n)$$

then

$$q_n = P(U_m \geq F(g_1 + n^{-1/2} a \log n))$$

where $U_{1n} < U_{2n} < \dots < U_{nn}$ are the order statistics of a sample from the uniform distribution on $(0,1)$. It is well known (see e.g. Breiman (1968), p. 285) that the distribution of U_m is the same as that of S_r/S_{n+1} where $S_j = Z_1 + \dots + Z_j$ with $\{Z_j\}$ independent random variables each exponentially distributed with parameter one. Applying Bernstein's inequality we have

$$q_n = P(S_r \geq b_n S_{n+1}) \leq E e^{t(S_r - b_n S_{n+1})}, \quad t > 0,$$

where $b_n = F(g_1 + n^{-1/2} a \log n)$. Choosing

$$t = \{(n+1)b_n - r\}((n+1)b_n(1 - b_n))^{-1}$$

we obtain $q_n = O(e^{-1/2 a^2 f_1^2 (\log n)^2 / \alpha \lambda})$.

Since the same result holds for $P(n^{1/2}(X_m - g_1) \leq -a \log n)$, (3) follows. (4) is proved in a similar way.

Let

$$(5) \quad Z_m = n^{1/2}(X_m - g_1)f_1(\alpha \lambda)^{-1/2}$$

and denote its density by $f_m(u)$. It is known (see e.g. Cramer (1966), p. 369) that for fixed u , $f_m(u) \rightarrow \phi(u)$, the standard normal density. The following result gives an L_1 rate for this convergence.

Lemma 2

$$(6) \quad I_n = \int_{-\infty}^{\infty} |f_m(u) - \phi(u)| du = O(n^{-1/2}).$$

Proof

We have

$$\begin{aligned} I_n &= \int_{-c_n}^{c_n} |f_m(u) - \phi(u)| du + \int_{-\infty}^{-c_n} |f_m(u) - \phi(u)| du + \int_{c_n}^{\infty} |f_m(u) - \phi(u)| du \\ &= R_{1n} + R_{2n} + R_{3n} \quad (\text{say}) \end{aligned}$$

$$\text{Now } R_{2n} \leq \int_{-\infty}^{-c_n} f_m(u) du + \int_{-\infty}^{-c_n} \phi(u) du$$

The first term $O(n^{-1/2})$ by (3) and the second is $O(n^{-1/2})$ by using the inequality $\Phi(x) < |x|^{-1}\phi(x)$ for $x < 0$. Thus $R_{2n} = O(n^{-1/2})$ and in the same way $R_{3n} = O(n^{-1/2})$.

We now consider R_{1n} . To shorten the proof somewhat, we take αn to be an integer; thus $r = \alpha n$. Only very minor modifications are needed when this is not the case. Put

$$(7) \quad t = g_1 + n^{-1/2} u (\alpha \lambda)^{1/2} / f_1$$

then

$$\begin{aligned} f_m(u) &= \binom{n}{r} F(t)^{r-1} (1 - F(t))^{n-r} f(t) n^{-1/2} (\alpha \lambda)^{1/2} / f_1 \\ &= (F(t)/\alpha)^r ((1 - F(t))/\lambda)^{n-r} (f(t)/f_1) \alpha F(t)^{-1} \\ &\quad \binom{n}{r} (n-r) \alpha^r \lambda^{n-r-1} n^{-1/2} (\alpha \lambda)^{1/2} \\ &= A_{1n}(u) A_{2n}(u) A_{3n} \quad (\text{say}) \end{aligned}$$

where

$$(8) \quad A_{1n}(u) = (F(t)/\alpha)^r ((1 - F(t))/\lambda)^{n-r}$$

$$(9) \quad A_{2n}(u) = (f(t)/f_1) \alpha F(t)^{-1}$$

$$(10) \quad A_{3n} = \binom{n}{r} (n-r) \alpha^r \lambda^{n-r-1} n^{-1/2} (\alpha \lambda)^{1/2}$$

We now show that

$$(11) \quad R_{4n} = \int_{-c_n}^{c_n} |A_{1n}(u) - e^{-1/2} u^2| du = O(n^{-1/2})$$

R_{4n} can be written as

$$R_{4n} = \int_{-c_n}^{c_n} e^{-\frac{1}{2}u^2} |e^{\log A_{1n}(u) + \frac{1}{2}u^2} - 1| du$$

and

$$(12) \quad \log A_{1n}(u) = r \log(1 + g(t)) + (n - r) \log(1 - h(t))$$

where

$$(13) \quad g(t) = (F(t) - \alpha)/\alpha; \quad h(t) = (F(t) - \alpha)/\lambda.$$

From (2), (7) and (13) it follows that

$$|g(t)| \leq \frac{1}{2} \text{ for } |u| < c_n$$

In this case we can write

$$(14) \quad \log(1 + g(t)) = g(t) - \frac{1}{2}g(t)^2 + \sum_{k=3}^{\infty} (-1)^{k-1} \frac{1}{k} g(t)^k$$

The last term can be bounded by

$$\begin{aligned} (15) \quad \frac{1}{3} \sum_k |g(t)|^k &= \frac{1}{3} |g(t)|^3 (1 - |g(t)|)^{-1} \\ &\leq \frac{1}{3} \alpha^{-3} |F(t) - \alpha|^3 \cdot 2 \\ &\leq \alpha^{-3} |F(t) - \alpha|^3 \end{aligned}$$

Applying Taylor's theorem we can write

$$(16) \quad F(t) - \alpha = n^{-\frac{1}{2}} u(\alpha\lambda)^{\frac{1}{2}} f_1^{-1} f(u_{1n})$$

with u_{1n} defined in the usual way.

Substituting (16) in (15) we get

$$(17) \quad r \left| \sum_{k=3}^{\infty} (-1)^{k-1} \frac{1}{k} g(t)^k \right| \leq n^{-\frac{1}{2}} |u|^3 d_1$$

with

$$(18) \quad d_1 = \alpha^{-2} f_1^{-3} ((\alpha\lambda)^{\frac{1}{2}} \|f\|)^3$$

For $|u| < c_n$ we also have $|h(t)| \leq \frac{1}{2}$ and we can write

$$(19) \log(1 - h(t)) = -h(t) - \frac{1}{2}h(t)^2 - \sum_{k=3}^{\infty} k^{-1}h(t)^k$$

As above, we find that for $|u| < c_n$

$$(20) (n-r)|\sum_{k=3}^{\infty} k^{-1}h(t)^k| \leq n^{-\frac{1}{2}}|u|^3 d_2$$

with

$$(21) d_2 = \lambda^{-2} f_1^{-3} ((\alpha\lambda)^{\frac{1}{2}} \|f\|)^3$$

Furthermore

$$(22) \begin{aligned} & r(g(t) - \frac{1}{2}g(t)^2) - (n-r)(h(t) + \frac{1}{2}h(t)^2) + \frac{1}{2}u^2 \\ &= -(2\alpha\lambda)^{-1} n(F(t) - \alpha)^2 + \frac{1}{2}u^2 \end{aligned}$$

Applying Taylor's theorem to second order, we get

$$(23) F(t) - \alpha = n^{-\frac{1}{2}}u(\alpha\lambda)^{\frac{1}{2}} + \frac{1}{2}(nf_1^2)^{-1}u^2\alpha\lambda f'(u_{2n})$$

with u_{2n} defined in the usual way. Using (23) the right-hand side of (22) becomes

$$-(2\alpha\lambda)^{-1} n(F(t) - \alpha)^2 + \frac{1}{2}u^2 = -\frac{1}{2}n^{-\frac{1}{2}}u^3(\alpha\lambda)^{\frac{1}{2}}f'(u_{2n})f_1^{-2} - 1/8 n^{-1}u^4\alpha\lambda f'(u_{2n})^2 f_1^{-4}$$

For $|u| < c_n$ we bound this by

$$(24) |-(2\alpha\lambda)^{-1} n(F(t) - \alpha)^2 + \frac{1}{2}u^2| \leq n^{-\frac{1}{2}}|u|^3 d_3$$

with

$$(25) d_3 = \frac{1}{2}(\alpha\lambda)^{\frac{1}{2}}\|f'\|f_1^{-2} + 1/8 c_0\alpha\lambda\|f'\|^2 f_1^{-4}$$

Taking (17), (20) and (24) together, give

$$(26) |\log A_{1n}(u) + \frac{1}{2}u^2| \leq n^{-\frac{1}{2}}|u|^3 d \text{ for } |u| < c_n,$$

with $d = d_1 + d_2 + d_3$.

Using the inequality $|e^x - 1| \leq |x|e^{|x|}$, it follows that

$$\begin{aligned}
R_{4n} &\leq \int_{-c_n}^{c_n} e^{-\frac{1}{2}u^2} |\log A_{1n}(u) + \frac{1}{2}u^2| e^{|\log A_{1n}(u) + \frac{1}{2}u^2|} du \\
&\leq dn^{-\frac{1}{2}} \int_{-c_n}^{c_n} |u|^3 e^{-\frac{1}{2}u^2} e^{n^{-\frac{1}{2}}|u|^3} du \\
&\leq dn^{-\frac{1}{2}} \int_{-c_n}^{c_n} |u|^3 e^{-\frac{1}{2}u^2} e^{cd^2} du \\
&\leq dn^{-\frac{1}{2}} \int_{-\infty}^{\infty} |u|^3 e^{-\frac{1}{2}u^2} (1 - 2cd) du
\end{aligned}$$

Choosing $c < (2d)^{-1}$, this term is $O(n^{-\frac{1}{2}})$ and thus $R_{4n} = O(n^{-\frac{1}{2}})$.

We now consider $A_{2n}(u)$. Applying Taylor's theorem, we have for $|u| < c_n$

$$\begin{aligned}
|A_{2n}(u) - 1| &= |(f(t)/f_1)\alpha F(t)^{-1} - 1| \\
&\leq \|f\| \|f_1^{-1}\| |F(t) - \alpha| F(t)^{-1} + n^{-\frac{1}{2}} |u| (\alpha\lambda)^{\frac{1}{2}} \|f'\| \|f_1^{-1}\|^2 \\
&\leq n^{-\frac{1}{2}} |u| \|f\|^2 \|f_1^{-1}\|^2 (\alpha\lambda)^{\frac{1}{2}} F(g_1 - c_0(\alpha\lambda)^{\frac{1}{2}}/f_1)^{-1} + n^{-\frac{1}{2}} |u| (\alpha\lambda)^{\frac{1}{2}} \|f'\| \|f_1^{-1}\|^2
\end{aligned}$$

Thus

$$(27) \quad |A_{2n}(u) - 1| \leq n^{-\frac{1}{2}} |u| b_1 \text{ for } |u| < c_n,$$

with the obvious expression for b_1 . From Cramer (1966), p. 200, we have that A_{3n} is bounded, say $A_{3n} \leq b_2$ and

$$(28) \quad |A_{3n} - (2\pi)^{-\frac{1}{2}}| = O(n^{-\frac{1}{2}})$$

We have

$$\begin{aligned}
(29) \quad A_{1n}(u)A_{2n}(u)A_{3n} - \phi(u) &= (A_{1n}(u) - e^{-\frac{1}{2}u^2})A_{2n}(u)A_{3n} \\
&\quad + e^{-\frac{1}{2}u^2}(A_{2n}(u) - 1)A_{3n} + e^{-\frac{1}{2}u^2}(A_{3n} - (2\pi)^{-\frac{1}{2}})
\end{aligned}$$

From (28) it follows that

$$(30) \quad \int_{-c_n}^{c_n} e^{-\frac{1}{2}u^2} |A_{2n}(u) - 1| A_{3n} du \leq n^{-\frac{1}{2}} b_1 b_2 \int_{-\infty}^{\infty} |u| e^{-\frac{1}{2}u^2} du = O(n^{-\frac{1}{2}}),$$

and from (28)

$$(31) \int_{-\infty}^{\infty} e^{-\frac{1}{2}u^2} |A_{3n} - (2\pi)^{-\frac{1}{2}}| du = O(n^{-\frac{1}{2}}).$$

Also, from $R_{4n} = O(n^{-\frac{1}{2}})$, we have

$$\begin{aligned} (32) \quad & \int_{-c_n}^{c_n} |A_{1n}(u) - e^{-\frac{1}{2}u^2}| |A_{2n}(u)| A_{3n} du \\ & \leq \|f\| f_1^{-1} \alpha F(g_1 - c_0(\alpha\lambda)^{\frac{1}{2}}/f_1)^{-1} b_2 \int_{-c_n}^{c_n} |A_{1n}(u) - e^{-\frac{1}{2}u^2}| du \\ & = O(n^{-\frac{1}{2}}) \end{aligned}$$

From (29), (30), (31) and (32) we have $R_{1n} = O(n^{-\frac{1}{2}})$ and (6) follows.

Remarks

1. Lemma 2 implies a Berry-Esseen result for Z_m as follows:
Denote the df. of Z_m by F_m . Then

$$|F_m(x) - \Phi(x)| = \left| \int_{-\infty}^x (f_m(u) - \phi(u)) du \right| \leq \int_{-\infty}^{\infty} |f_m(u) - \phi(u)| du$$

This gives

$$(33) \quad \sup_x |F_m(x) - \Phi(x)| = O(n^{-\frac{1}{2}})$$

See e.g. Reiss (1974) for results of this type.

2. An analogous result can be proved for the joint density of X_m and X_{sn} .

For the remainder of this section we assume $\int_{g_1}^{\infty} y^2 dF(y) < \infty$.

Define

$$(34) \quad \theta_n = (1 - F(t))^{-1} \int_t^{\infty} y f(y) dy$$

$$(35) \quad \theta = \lambda^{-1} \int_{g_1}^{\infty} y f(y) dy$$

with t as in (7). Note that for u fixed, $\theta_n \rightarrow \theta$ as $n \rightarrow \infty$. The next result gives a rate for this convergence.

Lemma 3

There is a constant K such that for $|u| \leq c_n$

$$(36) \quad |n^{1/2}(\theta_n - \theta) + \lambda^{-1}u(\alpha\lambda)^{1/2}(g_1 - \theta)| \leq n^{-1/2}Ku^2$$

Proof

Suppose first that $u > 0$. Then $t > g_1$ and

$$\begin{aligned} n^{1/2}(\theta_n - \theta) &= n^{1/2}[(1 - F(t))^{-1} - \lambda^{-1}] \int_{g_1}^{\infty} yf(y)dy - (1 - F(t))^{-1} \int_{g_1}^t yf(y)dy \\ &= n^{1/2}\lambda^{-1}(1 - F(t))^{-1}(F(t) - \alpha) \int_{g_1}^{\infty} yf(y)dy - n^{1/2}(1 - F(t))^{-1} \int_{g_1}^t yf(y)dy \end{aligned}$$

This gives

$$\begin{aligned} &n^{1/2}(\theta_n - \theta) + \lambda^{-1}u(\alpha\lambda)^{1/2}(g_1 - \theta) \\ &= (n^{1/2}(1 - F(t))^{-1}(F(t) - \alpha)\theta - \lambda^{-1}(\alpha\lambda)^{1/2}u\theta) \\ &\quad + (\lambda^{-1}(\alpha\lambda)^{1/2}g_1u - n^{1/2}(1 - F(t))^{-1} \int_{g_1}^t yf(y)dy) \\ &= V_{1n}(u) + V_{2n}(u) \quad (\text{say}) \end{aligned}$$

Using (16) and (23) we find

$$\begin{aligned} |V_{1n}(u)| &= |\theta\lambda^{-1}[(n^{1/2}(F(t) - \alpha) - (\alpha\lambda)^{1/2}u) + n^{1/2}(1 - F(t))^{-1}(F(t) - \alpha)^2]| \\ &\leq |\theta|\lambda^{-1}(\frac{1}{2}n^{-1/2}\alpha\lambda f_1^{-2}||f'||u^2 + n^{-1/2}\alpha\lambda f_1^{-2}||f||^2(1 - F(g_1 + c_0(\alpha\lambda)^{1/2}/f_1))^{-1}u^2) \end{aligned}$$

With the obvious expression for B_1 we have

$$(37) \quad |V_{1n}(u)| \leq n^{-1/2}B_1u^2$$

Consider $V_{2n}(u)$:

$$\begin{aligned} V_{2n}(u) &= \lambda^{-1}((\alpha\lambda)^{1/2}g_1u - n^{1/2} \int_{g_1}^t yf(y)dy \\ &\quad + n^{1/2}(1 - F(t))^{-1}(F(t) - \alpha) \int_{g_1}^t yf(y)dy) \\ &= \lambda^{-1}(r_{1n}(u) + r_{2n}(u)) \quad (\text{say}) \end{aligned}$$

Using Taylor's theorem we have

$$\begin{aligned}\int_{g_1}^t y f(y) dy &= n^{-1/2} (\alpha \lambda)^{1/2} u f_1^{-1} t_n f(t_n) \\ &= n^{-1/2} (\alpha \lambda)^{1/2} u g_1 + n^{-1/2} (\alpha \lambda)^{1/2} u f_1^{-1} (t_n f(t_n) - g_1 f_1)\end{aligned}$$

with t_n defined in the usual way. We have

$$\begin{aligned}|t_n f(t_n) - g_1 f_1| &= |(t_n - g_1) f_1 + t_n (f(t_n) - f_1)| \\ &\leq n^{-1/2} |u| (\alpha \lambda)^{1/2} [1 + (|g_1| + c_0 (\alpha \lambda)^{1/2} f_1^{-1}) f_1^{-1} \|f'\|]\end{aligned}$$

Using this, (16) and the fact that

$$(1 - F(t))^{-1} \leq (1 - F(g_1 + c_0 (\alpha \lambda)^{1/2} / f_1))^{-1}$$

We conclude

$$(38) \quad |V_{2n}(u)| \leq n^{-1/2} B_2 u^2$$

for some constant B_2 .

For $u > 0$ the result follows from (37) and (38). For $u < 0$ the proof follows in a similar way. Thus (36) holds.

Define

$$(39) \quad \tau_n^2 = (1 - F(t))^{-1} \int_t^\infty y^2 f(y) dy - \theta_n^2$$

$$(40) \quad \tau^2 = \lambda^{-1} \int_{g_1}^\infty y^2 f(y) dy - \theta^2$$

Note again that $\tau_n^2 \rightarrow \tau^2$ as $n \rightarrow \infty$.

Lemma 4

Let $|u| \leq c_n$

(i) There is a constant L such that

$$(41) \quad |\tau_n^2 - \tau^2| \leq n^{-1/2} L |u|$$

(ii) If $c < \min(c_0 L^{-1} \tau^2)$ then

$$(42) \quad \tau_n^2 \geq (\tau^2 - cL) > 0$$

and there is a constant M such that

$$(43) \quad |\tau_n^{-1} - \tau^{-1}| \leq n^{-1/2} M |\mu|$$

Proof

(i) By lemma 3 there is a K_1 such that

$$(44) \quad |\theta_n - \theta| \leq n^{-1/2} K_1 |\mu|$$

We also have, with $a_0 = g_1 - c_0(\alpha\lambda)^{1/2}/f_1$

$$(45) \quad |\theta_n + \theta| \leq |\theta| + (1 - F(g_1 + c_0(\alpha\lambda)^{1/2}/f_1))^{-1} \int_{a_0}^{\infty} |y| f(y) dy$$

(44) and (45) implies that there is a K_2 such that

$$(46) \quad |\theta_n^2 - \theta^2| = |\theta_n - \theta| |\theta_n + \theta| \leq n^{-1/2} K_2 |\mu|$$

In essentially the same way that lemma 3 was proved we can show that there is a K_3 such that

$$(47) \quad |(1 - F(t))^{-1} \int_t^{\infty} y^2 f(y) dy - \lambda^{-1} \int_{g_1}^{\infty} y^2 f(y) dy| \leq n^{-1/2} K_3 |\mu|$$

(46) and (47) yield the result.

(ii) The first part follows directly from (i). For the second part we have

$$|\tau_n^{-1} - \tau^{-1}| = \tau^{-1} \tau_n^{-1} (\tau_n + \tau)^{-1} |\tau_n^2 - \tau^2|$$

and the result follows from (i) and the first part of (ii).

The following is a slight extension of (36).

Lemma 5

Let $c < \min(c_0, L^{-1} \tau^2)$ and $|\mu| \leq c_n$; then there is a constant N such that

$$(48) \quad |n^{1/2}(\theta_n - \theta)\tau_n^{-1} + \lambda^{-1}u(\alpha\lambda)^{1/2}(g_1 - \theta)\tau^{-1}| \leq n^{-1/2}Nu^2$$

Proof

It follows directly from lemmas 3 and 4.

We need one more result. Define

$$(49) \quad \sigma_1^2 = \lambda^{-2} [\int_{g_1}^{\infty} (x - \theta)^2 dF(x) + \alpha\lambda(\theta - g_1)^2]$$

$$(50) \quad \alpha_n(x) = (\lambda^{1/2} \sigma_1 x - (\lambda n)^{1/2}(\theta_n - \theta))\tau_n^{-1}$$

$$(51) \quad \alpha(x) = (\lambda^{1/2} \sigma_1 x - \alpha^{1/2}u(g_1 - \theta))\tau^{-1}$$

$$(52) \quad \beta_n(x) = \tau\alpha(x)\tau_n^{-1}$$

Note that $\alpha_n(x)$, $\alpha(x)$ and $\beta_n(x)$ depend on u . For u fixed we have $\alpha_n(x) \rightarrow \alpha(x)$.

Lemma 6

If $c < \min(c_0, \tau^2 L^{-1}, \tau^{-1} M^{-1})$ and $|u| \leq c_n$ then there are constants P_1 and P_2 such that

$$(53) \quad \sup_x |\Phi(\alpha_n(x)) - \Phi(\alpha(x))| \leq n^{-1/2}(P_1|u| + P_2u^2).$$

Proof

Using Taylor's theorem and lemmas 3 and 4 it follows that

$$(54) \quad |\Phi(\alpha_n(x)) - \Phi(\beta_n(x))| \leq n^{-1/2}K(2\pi)^{-1/2}(\tau^2 - cL)^{-1/2}u^2$$

There exists a $0 < \delta < 1$, with $\delta = \delta_n(u, x)$, such that

$$\begin{aligned} |\Phi(\beta_n(x)) - \Phi(\alpha(x))| &= |\beta_n(x) - \alpha(x)| \phi(\delta\beta_n(x) + (1 - \delta)\alpha(x)) \\ &\leq \lambda^{1/2} \sigma_1 |x| |\tau^{-1} - \tau_n^{-1}| \phi(\delta\beta_n(x) + (1 - \delta)\alpha(x)) \\ &\quad + \alpha^{1/2} |u| |g_1 - \theta| (2\pi)^{-1/2} |\tau^{-1} - \tau_n^{-1}| \\ &\leq \lambda^{1/2} \sigma_1 M n^{-1/2} |u| |x| \phi(\delta\beta_n(x) + (1 - \delta)\alpha(x)) \\ &\quad + \alpha^{1/2} |g_1 - \theta| (2\pi)^{-1/2} n^{-1/2} M u^2 \end{aligned}$$

Using (43) we can write

$$\tau_n^{-1} = \tau^{-1} + r_n$$

with

$$(55) \quad |r_n| \leq n^{-1/2} M |u| \leq cM < \tau^{-1}$$

Since

$$\beta_n(x) = \tau \alpha(x) \tau_n^{-1} = \alpha(x)(1 + \tau r_n)$$

we have

$$(56) \quad \begin{aligned} \delta \beta_n(x) + (1 - \delta) \alpha(x) &= \delta \alpha(x)(1 + \tau r_n) + (1 - \delta) \alpha(x) \\ &= \alpha(x)(1 + \delta \tau r_n) \end{aligned}$$

Using (55) we have $\delta \tau |r_n| < 1$ and

$$\begin{aligned} |1 + \delta \tau r_n| &\geq 1 - \delta \tau |r_n| \\ &\geq 1 - \tau M c > 0 \end{aligned}$$

and thus

$$(1 + \delta \tau r_n)^2 \geq (1 - \tau M c)^2$$

This gives

$$\phi(\alpha(x)(1 + \delta \tau r_n)) \leq \phi(\alpha(x)(1 - \tau M c))$$

This with (56) yield

$$\sup_x |x| \phi(\delta \beta_n(x) + (1 - \delta) \alpha(x)) \leq \sup_x |x| \phi(\alpha(x)(1 - \tau M c))$$

which is finite. This implies the existence of a P such that

$$(57) \quad \sup_x |\Phi(\beta_n(x)) - \Phi(\alpha(x))| \leq n^{-1/2} P |u| + n^{-1/2} \alpha^{1/2} |g_1 - \theta| (2\pi)^{-1/2} M u^2$$

Using (54) and (57) the lemma follows from

$$\begin{aligned} \sup_x |\Phi(\alpha_n(x)) - \Phi(\alpha(x))| &\leq \sup_x |\Phi(\alpha_n(x)) - \Phi(\beta_n(x))| \\ &\quad + \sup_x |\Phi(\alpha(x)) - \Phi(\beta_n(x))| \end{aligned}$$

3. Main Results

We now present the main results. Define

$$(58) \quad \mu = \gamma^{-1} \int_{g_1}^{g_2} y dF(y)$$

$$(59) \quad \sigma^2 = \gamma^{-2} \left[\int_{g_1}^{g_2} (y - \mu)^2 dF(y) + \alpha(\mu - g_1)^2 + (1 - \alpha)(\mu - g_2)^2 - (\alpha(\mu - g_1) - (1 - \alpha)(g_2 - \mu))^2 \right]$$

where we again assume $0 < \alpha < \frac{1}{2}$.

Theorem 1

If $F_n(x) = P(n^{1/2}(M_n - \mu)/\sigma \leq x)$ then

$$(60) \quad \sup_x |F_n(x) - \Phi(x)| = O(n^{-1/2}(\log n)^2)$$

Proof

From the proof of theorem 1 of De Wet and Venter (1974) it follows that (using their notation for $\{Y_i\}$, V_{1n} , V_{2n} , A_n and B_n)

$$M_n - \mu = n^{-1} \sum_{i=1}^n Y_i + \gamma^{-1} n^{-1/2} (R_{5n} + R_{6n})$$

with

$$R_{5n} = n^{-1/2} (V_{1n} - g_1 A_n + \alpha g_1)$$

$$R_{6n} = n^{-1/2} (V_{2n} - g_2 B_n + \alpha g_2)$$

Since

$$(A_n - r)X_{r+1,n} \leq V_{1n} \leq (A_n - r)g_1$$

it follows that

$$n^{-1/2}g_1(\alpha n - r) + n^{-1/2}(X_{r+1,n} - g_1)(A_n - r) \leq R_{5n} \leq n^{-1/2}g_1(\alpha n - r)$$

This gives

$$|R_{5n}| \leq n^{-1/2}|g_1|(\alpha n - r) + n^{-1/2}|X_{r+1,n} - g_1||A_n - r|$$

Let $a_n = n^{-1/2}(\log n)^2$ and $\epsilon > 0$. Then for n large enough

$$\begin{aligned} P(|R_{5n}| \geq \epsilon a_n) &\leq P(|X_{r+1,n} - g_1||A_n - r| \geq \epsilon(\log n)^2 - |g_1|(\alpha n - r)) \\ &\leq P(n^{1/2}|X_{r+1,n} - g_1| \geq \epsilon \log n(1 - |g_1|(\alpha n - r)(\log n)^{-2})) \\ &\quad + P(n^{-1/2}|A_n - r| \geq \log n) \end{aligned}$$

The first term is $O(n^{-1/2})$ by lemma 1 while the second term is $O(n^{-1/2})$ by an easy application of the Berry-Esseen theorem for Binomial random variables. This gives

$$(61) \quad P(|R_{5n}| \geq \epsilon a_n) = O(n^{-1/2})$$

and in the same way it follows that

$$(62) \quad P(|R_{6n}| \geq \epsilon a_n) = O(n^{-1/2}).$$

Applying the Berry-Esseen theorem to $n^{-1}\Sigma Y_i$ we find that there is an A_1 such that

$$\begin{aligned} (63) \quad P(n^{-1/2}\Sigma Y_i \leq \sigma(x + a_n)) &\leq \Phi(x + a_n) + A_1 n^{-1/2} \\ &\leq \Phi(x) + (2\pi)^{-1/2}a_n + A_1 n^{-1/2} \\ &= \Phi(x) + r_n \end{aligned}$$

with $r_n = O(n^{-1/2}(\log n)^2)$ independent of x . In the same way

$$(64) \quad P(n^{-1/2}\Sigma Y_i \leq \sigma(x - a_n)) \geq \Phi(x) - r_n$$

The theorem now follows from (61), (62) and (64) and the inequalities

$$F_n(x) \leq P(n^{-1/2}\Sigma Y_i \leq \sigma(x + a_n)) + P(R_{5n} \leq -1/2\gamma\sigma a_n) + P(R_{6n} \leq -1/2\gamma\sigma a_n)$$

and

$$F_n(x) \geq P(n^{-1/2} \sum Y_i \leq \sigma(x - a_n)) - P(R_{5n} > \frac{1}{2} \gamma \sigma a_n) - P(R_{6n} > \frac{1}{2} \gamma \sigma a_n).$$

Remark

The $(\log n)^2$ can be improved to $(\log n)^{1+\epsilon}$ for $\epsilon > 0$. It does not seem possible to improve much on this by the above methods. To obtain a rate of $n^{-1/2}$ we use a different approach. To avoid an undue long proof, we only consider the one-sided trimmed mean, viz.

$$(65) \quad M_{1n} = \lambda^{-1} n^{-1} \sum_{i=r+1}^n X_{in}$$

with $r = [\alpha n]$ where in this case $0 < \alpha < 1$.

Define

$$F_{1n}(x) = P(n^{1/2}(M_{1n} - \theta)/\sigma_1 \leq x)$$

with θ, σ_1^2 as in (35) and (49).

Theorem 2

If $\int_1^\infty |y|^3 dF(y) < \infty$ then

$$(66) \quad \sup_x |F_{1n}(x) - \Phi(x)| = O(n^{-1/2})$$

Proof

Define Z_{in} and t as in (5) and (7). Let $W_{1n}, W_{2n}, \dots, W_{n-r,n}$ be i.i.d. random variables with df.

$$G_n(y) = \begin{cases} (F(y) - F(t))(1 - F(t))^{-1} & y \geq t \\ 0 & y < t \end{cases}$$

and corresponding density

$$g_n(y) = \begin{cases} f(y)(1 - F(t))^{-1} & y \geq t \\ 0 & y < t \end{cases}$$

We then have

$$EW_{1n} = \theta_n; \text{var} W_{1n} = \tau_n^2$$

with θ_n and τ_n^2 as in (34), (39).

It is known that conditional on $Z_{rn} = u$, $X_{r+1,n}, \dots, X_{nn}$ are distributed as the order statistics of a sample of size $n-r$ from the df. $G_n(y)$. Hence, conditional on $Z_{rn} = u$, M_{1n} is distributed as $\lambda^{-1} n^{-1} \sum_{i=1}^{n-r} W_{in}$. For convenience we again assume αn to be an integer. Define

$$W_n = (\lambda n)^{-1} \sum_{i=1}^{\lambda n} W_{in}$$

Choose c as in lemma 6. Then

$$\begin{aligned} F_{1n}(x) &= \int_{-\infty}^{\infty} P(n^{1/2}(M_{1n} - \theta) \leq \sigma_1 x / Z_{rn} = u) f_{rn}(u) du \\ &= \int_{-\infty}^{\infty} P(n^{1/2}(W_n - \theta) \leq \sigma_1 x) f_{rn}(u) du \\ &= \int_{-c_n}^{c_n} P(n^{1/2}(W_n - \theta) \leq \sigma_1 x) f_{rn}(u) du + \int_{-\infty}^{-c_n} P(n^{1/2}(W_n - \theta) \leq \sigma_1 x) f_{rn}(u) du \\ &\quad + \int_{c_n}^{\infty} P(n^{1/2}(W_n - \theta) \leq \sigma_1 x) f_{rn}(u) du \\ &= F_{1n}^*(x) + R_{7n} + R_{8n} \quad (\text{say}) \end{aligned}$$

By lemma 1

$$R_{7n} \leq \int_{-\infty}^{-c_n} f_{rn}(u) du = O(n^{-1/2})$$

and

$$R_{8n} = O(n^{-1/2}), \text{ independent of } x.$$

With $\alpha(x)$ as in (51) we have

$$\begin{aligned} \Phi(x) &= \int_{-c_n}^{c_n} \Phi(\alpha(x)) \phi(u) du + \int_{-\infty}^{-c_n} \Phi(\alpha(x)) \phi(u) du + \int_{c_n}^{\infty} \Phi(\alpha(x)) \phi(u) du \\ &= \Phi_n(x) + R_{9n} + R_{10n} \quad (\text{say}) \end{aligned}$$

Using the inequalities $\Phi(x) < |x|^{-1} \phi(x)$ for $x < 0$ and $1 - \Phi(x) < x^{-1} \phi(x)$ for $x > 0$ we have $R_{9n} = O(n^{-1/2})$ and $R_{10n} = O(n^{-1/2})$ independent of x . Let $H_n(x)$ denote the df. of $(\lambda n)^{1/2}(W_n - \theta_n)/\tau_n$ and let $\alpha_n(x)$ be as in (50). Then

$$P(n^{1/2}(W_n - \theta) \leq \sigma_1 x) = H_n(\alpha_n(x))$$

Applying the Berry-Esseen theorem in the form given by Chung (1968), p. 206, there exists a constant A_0 such that

$$(67) \quad |H_n(\alpha_n(x)) - \Phi(\alpha_n(x))| \leq A_0 T_n(u)$$

with

$$\begin{aligned} T_n(u) &= \lambda n E[(\lambda n)^{-1/2} |W_{1n} - \theta_n| \tau_n^{-1}]^3 \\ &= (\lambda n)^{-1/2} \tau_n^{-3} E|W_{1n} - \theta_n|^3 \end{aligned}$$

For $|u| \leq c_n$ there exists, by assumption, a Q_1 such that

$$E|W_{1n} - \theta_n|^3 \leq Q_1$$

Hence by lemma 4 there is a Q such that

$$(68) \quad T_n(u) \leq n^{-1/2} Q \text{ for } |u| \leq c_n.$$

We have

$$\begin{aligned} F_{1n}(x) - \Phi(x) &= \int_{-\infty}^{\infty} H_n(\alpha_n(x)) f_m(u) du - \Phi(x) \\ &= \int_{-c_n}^{c_n} (H_n(\alpha_n(x)) - \Phi(\alpha_n(x))) f_m(u) du \\ &\quad + \int_{-c_n}^{c_n} (\Phi(\alpha_n(x)) - \Phi(\alpha(x))) \phi(u) du + \int_{-c_n}^{c_n} \Phi(\alpha(x)) (f_m(u) - \\ &\quad - \phi(u)) du \\ &\quad + \int_{-c_n}^{c_n} (\Phi(\alpha_n(x)) - \Phi(\alpha(x))) (f_m(u) - \phi(u)) du + O(n^{-1/2}) \end{aligned}$$

with the $O(n^{-1/2})$ term consisting of R_{in} , $i=7,8,9,10$. The first term is $O(n^{-1/2})$ by (67) and (68), the second term is $O(n^{-1/2})$ by lemma 6 and the third and fourth terms are $O(n^{-1/2})$ by lemma 2. This completes the proof.

Remark

To prove the theorem for M_n , straightforward but tedious modifications have to be

made to the above proof. In this case we choose c_0 so small that F'' is bounded on $(g_1 - c_0(\alpha\lambda)^{1/2}/f_1, g_1 + c_0(\alpha\lambda)^{1/2}/f_1)$ and on $(g_2 - c_0(\alpha\lambda)^{1/2}/f_2, g_2 + (\alpha\lambda)^{1/2}/f_2)$. The condition of theorem 2 then becomes

$$\int_{g_1}^{g_2} |y|^3 dF(y) < \infty,$$

which is superfluous and we obtain a rate of $O(n^{-1/2})$ under the same conditions as theorem 1. The proof of theorem 1 illustrates the relative ease with which a rate of $O(n^{-1/2}(\log n)^2)$ can be obtained compared to a rate of $O(n^{-1/2})$.

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